Di Yang

3D/4D Generative Modeling | **Learning-Based Robot Planning** | **Ph.D. Researcher**

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Research Profile

Ph.D. researcher working on 3D/4D generative modeling and learning-based robot planning, with applications in embodied AI.

Education

William & Mary — Williamsburg, VA
Ph.D. Student in Data Science | *Advisor: Dr. Yanhai Xiong* — *Sep 2024 - Expected Dec 2027*

National University of Singapore — Singapore
M.Sc. in Electrical Engineering | *Advisor: Dr. John S. Ho* — *Jan 2021 - Jan 2022*

University of Electronic Science and Technology of China — Chengdu, China
B.Eng. in Microelectronic Science and Engineering — *Sep 2016 - Jul 2020*

Selected Publications

Peer-reviewed publications

- **D. Yang** et al. "LENS: Learning to Navigate with Active Search for Partially Observable MAPF in Unknown Environments." *TMLR*, 2026.
- **D. Yang**, Y. Xiong. "UP-Bench: A Benchmark for Underwater Path Planning Algorithms." *KDD 2025 Benchmark Track* (**Oral**).
- Y. Li, **D. Yang***, H. Chen, Y. Xiong. "HyCO: A Hybrid Neural Solver for Combinatorial Optimization." **NeurIPS 2026** (Poster). (*Co-first author)
- H. Jiang, Y. Li, **D. Yang***, Y. Xiong, H. Chen, Y. He. "VTV-FM: Flow Matching through Variational Terminal-Velocity Closure." **NeurIPS 2026** (Poster). (*Co-first author)

Manuscripts

- **D. Yang**, Z. Li, Y. Xiong, Y. Wang. "Channel Match: Calibrating Shape Latents for Few-Step 3D Generation." Under review at **AAAI 2027**.
- **D. Yang**, Z. Li, Y. Xiong, Y. Wang. "A Compact Explicit 4D Representation for Dynamic Scenes." Preprint, 2026.

Selected Research Experience

SparcAI Inc. — Jan 2026 - Present
Research Collaboration and Research Intern | *Advisor: Dr. Yufei Wang* — *Remote*

Compact 4D Representation Learning for Dynamic Scenes

- Developed a feed-forward autoencoder that encodes posed monocular video into shared static features and spatially anchored temporal slots, decoded as explicit **2D Gaussian surfels**.
- In the latest iteration, reduced time-varying feature storage by a median **5.0x** with negligible reconstruction change; the complete 32-frame state averages **0.67M 32-bit-equivalent values** on random windows.
- Adapted the scene encoder to ManiSkill3 and trained a flow-matching manipulation policy on frozen static and per-frame dynamic features, reaching **81.3% StackCube success** with 1,000 demonstrations.

Channel Match: Efficient Few-Step Generative Modeling

- Distilled a many-step native 3D generator into a **two-step** student using continuous self-consistency training, achieving **4.5x faster** shape sampling.
- Implemented a custom **JVP FlashAttention kernel** for forward-mode differentiation over sparse structured 3D latents.
- Identified channel-wise latent drift affecting downstream texture synthesis; proposed a training-free gated calibration without model retraining.

William & Mary — Sep 2024 - Present
Ph.D. Researcher | *Advisor: Dr. Yanhai Xiong* — *Williamsburg, VA*

LENS: Learning-Based Planning under Partial Observability

- Proposed a hierarchical multi-agent planner separating macro-topological search from local collision avoidance in unknown environments.
- Matched the OOD generalization of an 85M-parameter MAPF foundation model using **less than 0.2%** of its training data.

Additional Research

William & Mary Sep 2024 - Present
Ph.D. Researcher | Advisor: Dr. Yanhai Xiong *Williamsburg, VA*

HyCO: Hybrid Reinforcement Learning and Diffusion for Optimization

- Designed an RL-to-diffusion handover using policy entropy and model disagreement to combine sequential construction with generative solution completion.
- Reduced TSP-1000 optimality gap from **15.4% to 4.6%** over the RL backbone; matched a 16-sample diffusion baseline on TSP-100 with one guided completion.

UP-Bench: 3D Underwater Planning Benchmark

- Designed and open-sourced **UP-Bench** (KDD 2025 Benchmark Track, **Oral**), a 3D underwater simulator with fluid dynamics and dynamic obstacles; benchmarked classical planners and SAC agents.

VTV-FM: Second-Order Flow Matching

- Co-developed phase-space flow matching with cubic Hermite trajectories and a closed-form minimum-acceleration variational closure.
- Evaluated across synthetic systems, PDE fields, and CIFAR-10, achieving **FID 2.99** versus 3.45-3.73 for compared baselines.

SparcAI Inc. Jan 2026 - Present
Research Collaboration and Research Intern | Advisor: Dr. Yufei Wang *Remote*

Multimodal Policy Learning for Vision-Language-Action Manipulation

- Integrating robot-centric point maps inspired by *See Like a Robot* into a **5B VLA policy** to incorporate metric spatial information into manipulation.
- Built OmniGibson/BEHAVIOR training and evaluation pipelines aligning RGB-D, robot state, language, and actions; developed a **failure-case-driven data augmentation pipeline**.
- Configured module-specific learning rates for downstream fine-tuning, using a lower rate for the pretrained VLM backbone than for the action DiT and projector.

Engineering Experience

Cambricon Technologies Co., Ltd. Aug 2022 - Jul 2024
Digital Logic Engineer

- Built functional verification pipelines for large-scale AI accelerator chips under timing and scalability constraints.
- Owned system-level performance verification for low-latency data flow and error-handling paths.

Technical Skills

Machine learning: multimodal policy fine-tuning, reinforcement learning (SAC, PPO), imitation learning, diffusion, flow matching, consistency distillation, representation learning
Implementation: Python, PyTorch, Hugging Face, Triton, distributed training (DDP/multi-GPU), sparse convolution, gsplat, Linux
Data & evaluation: RGB-D/video pipelines, failure-case data augmentation, simulation benchmarks, OmniGibson, BEHAVIOR-1K, HoloOcean

Honors

Summer Research Fellowship, William & Mary (2025, 2026)
Scholarship for Model Student, University of Electronic Science and Technology of China (2018, 2019)